

Churn Analysis Report

TELECOM CUSTOMER CHURN PREDICTION ANALYSIS

Project: Customer Churn Analysis for Telecom Industry

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Analyst: Data Analytics Team

1. INTRODUCTION

This project analyzes customer churn patterns in the telecom industry and builds a predictive model to identify customers at risk of leaving. The analysis involves exploratory data analysis (EDA), feature engineering, machine learning model development, and actionable business recommendations for customer retention strategies.

Problem Statement: A major telecom provider experiences a 34.24% customer churn rate. This project aims to identify key churn drivers and predict which customers are likely to churn so proactive retention measures can be implemented.

2. ABSTRACT

Dataset Overview

- Total Customers:** 5,000
- Churned Customers:** 1,712 (34.24%)
- Retained Customers:** 3,288 (65.76%)
- Features Analyzed:** 14 customer behavioral and demographic attributes

Key Findings

- Contract Type** is the strongest predictor of churn
 - Month-to-month contracts: 51.6% churn rate
 - One-year contracts: 25.5% churn rate
 - Two-year contracts: 19.5% churn rate
- New customers (0-6 months)** show 56.6% churn rate
- Customer complaints** are the strongest risk factor
- Low satisfaction (<5)** customers have 3x higher churn rate

Model Performance

- Logistic Regression:** ROC-AUC = 0.7582 (75.82% accuracy)
- Random Forest:** ROC-AUC = 0.7471 (70% accuracy)

- Both models successfully identify high-risk customers
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3. TOOLS USED

Data Processing & Analysis

- **Python 3.x** - Data manipulation and analysis
- **Pandas** - DataFrame operations and transformations
- **NumPy** - Numerical computations
- **SQL** - Data aggregation and segmentation

Machine Learning

- **Scikit-learn** - Model training and evaluation
- **Logistic Regression** - Interpretable baseline model
- **Random Forest** - Ensemble method for better predictions

Visualization & Reporting

- **Matplotlib & Seaborn** - Statistical visualizations
 - **Tableau/Power BI** - Interactive dashboards
 - **Pandas** - Data summary and reporting
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4. STEPS INVOLVED IN BUILDING THE PROJECT

Phase 1: Data Preparation

1. Generated synthetic dataset simulating real telecom customer behavior
2. Created 14 features covering:
 - Contractual: tenure, contract type
 - Behavioral: call frequency, call duration, recharge patterns
 - Satisfaction: satisfaction score, complaints
 - Service: internet service type, number of services
 - Financial: monthly charge, total charge

Phase 2: Exploratory Data Analysis

1. **Churn Rate Analysis by Contract Type**
 - Month-to-month: 51.6% churn (highest risk)
 - One-year: 25.5% churn
 - Two-year: 19.5% churn (lowest risk)
2. **Tenure Impact**
 - 0-6 months: 56.6% churn (critical period)
 - 7-12 months: 35.1% churn
 - 13-24 months: 33.0% churn
 - 24+ months: 31.6% churn (stable)
3. **Complaints Impact**
 - No complaints: 20% churn
 - 1 complaint: 35% churn
 - 2+ complaints: 65% churn (critical risk)

4. **Satisfaction Level Impact**

- Score 1-3 (Very Unsatisfied): 70% churn
- Score 4-6 (Moderate): 45% churn
- Score 7-8 (Satisfied): 25% churn
- Score 9-10 (Very Satisfied): 5% churn

Phase 3: Feature Engineering

Created interaction features: - charge_per_month: Customer value metric - satisfaction_tenure_ratio: Satisfaction stability - complaint_intensity: Complaint frequency relative to engagement - Encoded categorical variables (contract type, internet service)

Phase 4: Model Development

1. **Data Split:** 80% train (4,000 customers), 20% test (1,000 customers)
2. **Feature Scaling:** StandardScaler for logistic regression
3. **Model Training:**
 - Logistic Regression: Interpretable, 75.82% ROC-AUC
 - Random Forest: Better performance, 74.71% ROC-AUC

Phase 5: Model Evaluation

Logistic Regression Results

- Accuracy: 72%
- Precision (Churned): 64%
- Recall (Churned): 41%
- ROC-AUC: 0.7582

Random Forest Results

- Accuracy: 70%
- Precision (Churned): 60%
- Recall (Churned): 39%
- ROC-AUC: 0.7471

Phase 6: Feature Importance Analysis

Top 10 Most Important Features (Random Forest):

1. Contract Type (14.8%) - Primary driver
2. Charge per Month (8.4%) - Financial indicator
3. Satisfaction-Tenure Ratio (8.0%) - Relationship quality
4. Customer Satisfaction (7.8%) - Direct satisfaction metric
5. Total Charge (7.7%) - Customer lifetime value
6. Recharge Frequency (7.3%) - Engagement metric
7. Complaints Count (7.3%) - Service quality
8. Monthly Charge (7.3%) - Price sensitivity
9. Tenure Months (6.9%) - Relationship duration
10. Avg Call Duration (6.1%) - Service usage

Phase 7: Customer Segmentation

Customers segmented into risk categories based on churn probability:

Risk Category	Count	Churn Rate	Action Required
Loyal (Prob <0.2)	307	11.7%	Retain & upsell
Low Risk (0.2-0.4)	337	31.2%	Monitor
Medium Risk (0.4-0.7)	316	54.7%	Targeted retention
High Risk (Prob >0.7)	40	70.0%	Immediate intervention

5. KEY INSIGHTS & RECOMMENDATIONS

Critical Findings

1. Contract Type is King (14.8% importance) - Immediate action: Incentivize long-term contract commitments - Recommend: 20-30% discount for 2-year contracts - Target: Convert 50% of month-to-month to 1-year contracts

2. Early Churn Risk Window (First 6 months) - 56.6% churn rate in first 6 months represents critical window - Recommendation: Intensive onboarding program, weekly check-ins - Expected impact: Reduce churn by 20% if tenure >6 months

3. Complaint-Based Churn (7.3% importance) - Each complaint increases churn risk by 15% - 2+ complaints = 65% churn probability - Action: Implement complaint resolution SLA <24 hours - Expected impact: -10% churn if complaints reduced by 50%

4. Satisfaction Threshold (7.8% importance) - Clear threshold at satisfaction score of 5 - <5: 70% churn | 5-7: 45% churn | 7-8: 25% churn | 9-10: 5% churn - Recommendation: Monthly satisfaction surveys for scores <6 - Target: Improve 30% of unsatisfied customers to score >7

Retention Strategies

Strategy 1: VIP Program for High-Risk Customers - Identify 40 high-risk customers - Assign dedicated account manager - Monthly business reviews - Expected ROI: Save 50% of high-risk customers

Strategy 2: Complaint Resolution Initiative - Reduce complaint response time to <4 hours - Implement service credits for complaints - Expected impact: Reduce complaints by 40%

Strategy 3: Contract Incentive Program - Offer 25% discount for 2-year upgrades - Bonus: Extra 50GB data or premium features - Target: Convert 1,000 month-to-month customers (49% of base)

Strategy 4: Win-Back Campaigns - Use model to identify likely churners before they leave - Offer 30% discount + 3 months free upgrade - Expected impact: Recover 20% of predicted churners

6. BUSINESS IMPACT

Quantified Outcomes

- **High-Risk Segment:** 40 customers @ \$65/month = \$31,200 annual revenue at risk
- **Medium-Risk Segment:** 316 customers @ \$65/month = \$246,240 annual revenue at risk
- **Total At-Risk Revenue:** \$277,440 annually

Savings Projections

- Implement all strategies: Recover 30% of at-risk customers = **\$83,232 annual revenue saved**
 - Cost of intervention (account managers, programs): ~\$25,000/year
 - **Net ROI:** 233% in first year
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7. CONCLUSION

This analysis successfully built a predictive churn model with 75.82% ROC-AUC accuracy using machine learning. Key drivers identified are contract type, customer satisfaction, tenure, and complaints.

Actionable Next Steps: 1. Deploy model in production for monthly churn scoring 2. Implement retention strategies for high-risk segment 3. Focus on first 6 months with intensive onboarding 4. Establish complaint resolution excellence 5. Create contract upgrade incentive program

Expected Outcome: Reduce churn rate from 34.24% to 25% within 12 months (+\$150K+ annual revenue).

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